

Flat Modules and Tor: Exercises

Sean Kim

Math 121

July 26, 2026

Problem 1. (*Proving Proposition 3*) Let R be a ring and M an R -module.

- a) Let $I \subset R$ be an ideal. Prove that $I \otimes_R M \rightarrow M$ is an injection iff $\mathrm{Tor}_1^R(R/I, M) = 0$ by considering the long exact sequence of Tor derived from the short exact sequence $0 \rightarrow I \rightarrow R \rightarrow R/I \rightarrow 0$. Then show that when M is flat, this condition is satisfied for all finitely generated ideals $I \subset R$.
- b) Suppose that the injectivity condition holds for all finitely generated ideals I . Show that for an arbitrary ideal $I' \subset R$ that is not necessarily finitely generated, the multiplication map $I' \otimes_R M \rightarrow M$ is injective.
- c) Let $N' \subset N$ be an inclusion of R -modules. To prove the injectivity of $N' \otimes_R M \rightarrow N \otimes_R M$, show that it suffices to consider the case where N is finitely generated and N/N' is cyclic.
- d) Recall from the modules worksheet that every cyclic R -module is isomorphic to R/I for some ideal $I \subset R$. Assuming $N/N' \cong R/I$, use the long exact sequence of Tor derived from $0 \rightarrow N' \rightarrow N \rightarrow N/N' \rightarrow 0$ to show that $N' \otimes_R M \rightarrow N \otimes_R M$ is injective. Conclude that M is flat and explain why we have just shown that M is flat iff the multiplication map $I \otimes_R M \rightarrow M$ is injective for all finitely generated ideals $I \subset R$.

Problem 2. (*Corollary 4 and Applications*) Let $(R, \mathfrak{m})$ be a local Noetherian ring, and let M be a finitely generated R -module.

- a) Prove that if M is flat, then M is free. (Hint: Choose elements $m_1, \dots, m_t \in M$ whose images form a basis of the vector space $M/\mathfrak{m}M$ over $R/\mathfrak{m}$. Then use Nakayama to obtain a surjection $R^n \rightarrow M$, and use flatness to force the kernel to vanish)
- b) Take $R = \mathbb{K}[x]_{\langle x \rangle}$ with the unique maximal ideal $\mathfrak{m} = \langle x \rangle$, and consider

$$M = R/\langle x \rangle \cong \mathbb{K}.$$

Prove that M is not flat over R by using the free resolution

$$0 \longrightarrow R \xrightarrow{\cdot x} R \longrightarrow R/\langle x \rangle \longrightarrow 0$$

to calculate $\mathrm{Tor}_1^R(R/\mathfrak{m}, M)$.

c) Let $R = \mathbb{Z}_{\langle p \rangle}$, the localization of $\mathbb{Z}$ at the prime ideal $\langle p \rangle$, and consider

$$M = R/\langle p \rangle \cong \mathbb{F}_p.$$

Show that M is not flat over R by calculating $\mathrm{Tor}_1^R(R/\langle p \rangle, M)$.

Problem 3. (*Computations with Tor*) Let $R = \mathbb{Z}$.

a) Use the free resolution

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z} \longrightarrow \mathbb{Z}/n\mathbb{Z} \longrightarrow 0$$

for $\mathbb{Z}/n\mathbb{Z}$ to compute $\mathrm{Tor}_1^{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, A)$ for any $\mathbb{Z}$ -module (abelian group) A . What happens if A is flat? Compare with Corollary 2.

b) Compute $\mathrm{Tor}_1^{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}/m\mathbb{Z})$.

c) Let R be a ring and suppose that $x \in R$ is a nonzerodivisor, with M an R -module. Calculate $\mathrm{Tor}_i^R(R/\langle x \rangle, M)$ by using the fact that

$$0 \longrightarrow R \xrightarrow{\cdot x} R \longrightarrow R/\langle x \rangle \longrightarrow 0$$

is a free resolution of $R/\langle x \rangle$. Compare with parts (a) and (b).

Problem 4. (*Applications of Corollary 2*) Let $R = \mathbb{Z}$. Which one of the following $\mathbb{Z}$ -modules are flat?

$$\mathbb{Z}, \quad \mathbb{Q}, \quad \mathbb{Z}/n\mathbb{Z}, \quad \mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z}, \quad \mathbb{Q}/\mathbb{Z}, \quad \mathbb{Z}_{\langle p \rangle}.$$

Problem 5. (*Applications of Corollary 3*) Consider the local Noetherian rings $(R, \mathfrak{m}) = (\mathbb{K}[x]_{\langle x \rangle}, \langle x \rangle)$ and $(S, \mathfrak{n}) = (\mathbb{K}[[x]], \langle x \rangle)$.

a) Explain how we can think about S as an R -algebra. Also show that $\mathfrak{m}S \subset \mathfrak{n}$.

b) Use the free resolution

$$0 \longrightarrow R \xrightarrow{\cdot x} R \longrightarrow R/\langle x \rangle \longrightarrow 0$$

to show that

$$\mathrm{Tor}_1^R(R/\langle x \rangle, S) \cong \ker(S \xrightarrow{\cdot x} S).$$

Then show that this kernel is 0, and deduce that $\mathbb{K}[[x]]$ is flat as a $\mathbb{K}[x]_{\langle x \rangle}$ -module.